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Creating an Intuitive Sound Design Application for a Wide Audience

My idea is to build an application that allows users to design a custom soundscape through a graphical interface. The soundscapes they create can be used to make loops, songs, MIDI instruments, and ultimately be shared in a social media context. While I want this app to be a useful tool for seasoned musicians, the intended audience is a younger, less experienced population that is used to consumer-creative apps like Instagram. The ultimate goal is to get that audience interested and involved in sound design, similar to how Instagram has gotten many young people into photography. I intend to do this by making the interface fun, intuitive and instantly rewarding.

The idea is that the user will draw (on a touchscreen) a top-down landscape using different materials. Behind the scenes, algorithms will procedurally use the landscape to modify sound waves. The user will constantly be given feedback so they can hear how their sound is developing as they are drawing. To keep visual learners (most people) engaged, the tools will produce vibrant drawings and ideally a dynamic drawing will produce an intriguing, unique sound. This is in contrast to the typical digital instrument editors, often designed to look like analog synths, which would scare away inexperienced and casual users. Once finished with their soundscape, the user can use a virtual MIDI pad (a grid where each cell is assigned to a user-created sound) to play their sounds in real time and can also record their performance. Tempo and volume can be controlled by simple sliders and also recorded live. The performance can then be shared on a social network where friends or followers can listen to, like, share, and if allowed, download your song or soundscape. This idea could be expanded upon massively, incorporating a MIDI note editor, stacked effects, presets, etc.

My goal for this project is to design and implement a well built, heavily tested and innovative interface for the graphical sound design component. I will need to do extensive research on acoustics and synthetically generated sound. I will experiment with different programming languages and interfaces, building the project incrementally, and will test builds on human subjects. Ultimately, I hope to carry out this project beyond 495 and 496 so I can implement the other essential features to make this a social media app.